WEEK 10 · LECTURE L3 · UNIT IV

Automation, Tasks and the Future of Work

Which work technology takes over, which it creates, and what a future of work HR strategy must cover.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Analyze work at the level of tasks with Autor, Levy and Murnane.
2. Explain why estimates of jobs at risk differ (Frey and Osborne; Arntz et al.).
3. Distinguish displacement from reinstatement (Acemoglu and Restrepo), and weigh Susskind’s task encroachment and frictions.
4. Set out the six elements of a future of work HR strategy.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 1 and CLO 4.

COMMIT BEFORE THEORY

Commit to an answer first

A report says 47% of jobs are at high risk of automation. What is the most careful reading?

A

Nearly half of all workers will lose their jobs soon

B

Many jobs contain tasks that could be automated; whether jobs vanish depends on which tasks, at what cost, and what new tasks appear

C

The figure is meaningless

D

Only manual jobs are at risk

EXAM LENS

The answer is B.

AUTOR, LEVY AND MURNANE (2003)

The task model

Routine cognitive

Rule based information tasks, such as data entry: substituted by computers.

Routine manual

Repetitive physical tasks, such as assembly: substituted.

Nonroutine analytic and interactive

Problem solving, persuasion, management: complemented.

Nonroutine manual

Tasks needing dexterity and adaptability, such as care or cleaning: hard to automate.

EXAM LENS

Technology substitutes for routine tasks and complements nonroutine ones. AI now reaches some nonroutine cognitive tasks, so the routine line keeps moving (Course framing). Analyze tasks, not job titles.

HOW MANY JOBS ARE AT RISK?

It depends on the unit

47%

of US jobs at high risk, judging whole occupations (Frey and Osborne, 2017).

About 9%

of jobs at high risk across OECD countries, judging tasks within jobs (Arntz, Gregory and Zierahn, 2016).

EXAM LENS

Plan for changing tasks within jobs, not only for jobs disappearing. “60% of the work” is not 60% of jobs.

ACEMOGLU AND RESTREPO (2019)

Displacement and reinstatement

Displacement

Automation takes over tasks labor used to do, reducing labor’s share.

Productivity

Cheaper production can raise demand and employment elsewhere.

Reinstatement

New tasks are created in which labor has an advantage.

EXAM LENS

The net effect depends on whether new tasks keep pace with automated ones. In recent decades they find displacement has outpaced reinstatement in the United States. A strategy should name the new tasks it creates, such as data quality ownership for displaced clerical staff.

SUSSKIND (2020)

Task encroachment, and the frictions

Machines take tasks, not jobs

Capabilities spread across tasks, including some once thought to need judgment; the share left to humans may shrink, reducing total demand for human work, not only shifting it.

Frictions slow the shift

Mismatches of skills, identity (reluctance to take work seen as lower status) and place keep people from moving to new work.

Policy matters

How income and meaning are shared becomes a central question.

EXAM LENS

Present Susskind as one side of a debate, with Acemoglu and Restrepo’s reinstatement effect as the counterpoint. Reskilling alone does not remove identity and place frictions.

WORLD ECONOMIC FORUM (2025)

The global picture to 2030

Jobs

About 170 million jobs created and 92 million displaced worldwide: a net gain of about 78 million.

Skills

Employers expect about 39% of workers’ core skills to change by 2030.

The caution

A net global gain does not mean no worker is harmed; displacement is concentrated in particular roles and places.

LEPAK AND SNELL (1999)

Match the employment mode to the value and uniqueness of skills

	Generic skills	Unique skills
High value	Acquire (buy)	Develop internally (build)
Low value	Contract (borrow)	Alliance

EXAM LENS

Skills move between quadrants as technology changes. Frey and Osborne had experts label 70 occupations and extended the estimates to 702 by model; reading their scores as exposure to change, not certainty of loss, fits this task based view.

THE 6B FRAMEWORK

Six workforce responses

Response	Meaning
Build	Reskill internally
Buy	Hire externally
Borrow	Contract or partner
Bridge	Move people to an adjacent role
Bot	Automate, with a transition plan for those affected
Exit	Phase out through natural attrition

COURSE FRAMING

The 6B framework is the course’s own, extending Lepak and Snell: Build for unique skills, Buy for valuable generic ones, Borrow for generic, low value work. Ground it in the architecture and in automation evidence when using it.

A FUTURE OF WORK HR STRATEGY

Six elements

Five questions it answers

Which work changes, and how fast? Who does it: employees, gig workers or machines? How do people move to new work? Is it fair to the least advantaged? How will HR govern it?

Task map

Which tasks in each role will be automated, augmented or created.

Reskilling and redeployment

Paths from shrinking tasks to growing ones, with time to learn.

Workforce mix

The core, contractors and platform workers, and duties to each.

Flexibility

Place and time, for the whole workforce.

Governance of algorithms

Audit, review and redress for automated decisions.

Measures

Leading indicators that show which future is arriving.

COURSE FRAMING

A strategy covers all six; a project list usually covers one or two.

APPLY

Automation in a shared services center

Teaching case (Course framing)

A Pune shared services center of 1,500 people processes invoices, payroll queries and onboarding documents. A vendor claims its AI can automate 60% of the work within two years. Staff are mostly graduates from nearby towns with ten or more years of service.

Work it through

1. Sort the center’s tasks with the Autor framework.
2. Is “60% of the work” the same as 60% of jobs?
3. Which new tasks could reinstate labor here?
4. Which of Susskind’s frictions matter most for these staff?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
High automation risk means the job will disappear	Most jobs mix tasks; tasks change, jobs are reshaped
Automation only displaces work	Productivity and reinstatement can create new tasks
Reskilling solves everything	Identity and place frictions also block transitions
47% and 9% cannot both be right	They measure different units: occupations against tasks

COMPARE AND EVALUATE

Five lenses on automation

Lens	Explains	Watch out for
Autor et al. (2003)	Which tasks change	The routine line moves
Frey and Osborne (2017)	Occupations at risk	Whole occupations overstate risk
Arntz et al. (2016)	Tasks within jobs	Lower, but still material
Acemoglu and Restrepo (2019)	Displacement against reinstatement	New tasks may lag
Susskind (2020)	Encroachment and frictions	One side of a debate

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse

An Indian bank expects automation to change half the tasks in its operations roles within five years. Using the task model, the displacement and reinstatement framework, and Susskind's frictions, design the elements of a future of work HR strategy. **20 marks**

A strong answer: analyzes roles by task; distinguishes displacement from reinstatement and names new tasks; addresses skills, identity and place frictions; covers workforce mix, flexibility, algorithm governance and measures, with a limit.

Explain why estimates of jobs at risk from automation range from about 9% to 47%, and what this means for HR planning. **20 marks**

A strong answer: contrasts occupation and task based methods; concludes for task level planning and reskilling.

Using Autor, Levy and Murnane (2003), explain why automation changes tasks rather than eliminating jobs. What is the main limitation of this view today? **10 marks**

A strong answer: explains the task model with an HR example, then names the moving routine line under AI.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Analyze tasks, not jobs.**
   Substitute routine, complement nonroutine.

2. **The unit decides the estimate.**
   Occupations overstate risk; tasks within jobs show change.

3. **Displacement and reinstatement both happen.**
   The net effect depends on new tasks keeping pace.

4. **A strategy covers six elements.**
   Frictions of skills, identity and place decide who is left behind.

SET READINGS · FROM THE LESSON PLAN

Read for this week

- Susskind, D. (2020). *A world without work*, selected chapter.
- Scully-Russ, E., & Torraco, R. J. (2020), whole article.

NEXT WEEK

- Westerman et al. (2014), leadership and strategy chapters; Scully-Russ and Torraco, pp. 84 to 85 and 88 to 89; sample Board memos.

> **NOTE**
> The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Lepak, D. P., & Snell, S. A. (1999). The human resource architecture. *Academy of Management Review, 24*(1), 31 to 48. https://doi.org/10.2307/259035
- World Economic Forum. (2025). *The future of jobs report 2025*. World Economic Forum.
- Acemoglu, D., & Restrepo, P. (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives, 33*(2), 3 to 30. https://doi.org/10.1257/jep.33.2.3
- Arntz, M., Gregory, T., & Zierahn, U. (2016). *The risk of automation for jobs in OECD countries* (OECD Social, Employment and Migration Working Papers No. 189). OECD Publishing.
- Autor, D. H., Levy, F., & Murnane, R. J. (2003). The skill content of recent technological change. *The Quarterly Journal of Economics, 118*(4), 1279 to 1333. https://doi.org/10.1162/003355303322552801
- Frey, C. B., & Osborne, M. A. (2017). The future of employment. *Technological Forecasting and Social Change, 114*, 254 to 280. https://doi.org/10.1016/j.techfore.2016.08.019
- Susskind, D. (2020). *A world without work*. Allen Lane.